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Determinants of a Matrix

A **determinant** is a scalar value that can be calculated from the elements of a square matrix.

It is an arrangement of numbers in the form $\begin{vmatrix} a & b & c \\ \dots & \dots & \dots \\ x & y & z \end{vmatrix}$

Determinant for a 3×3 matrix is determined by;

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2).$$

In this article, we come across properties of determinants, multiplication of determinants and determinants formula.

Table of Contents

- Introduction to Determinants
- SARRUS Method
- Symmetric and Skew Symmetric Determinants
- Multiplication of Two Determinants

Introduction to Determinants

The development of determinants took place when mathematicians were trying to solve a system of simultaneous linear equations (<https://byjus.com/jee/linear-equations/>).

$$E. g. \begin{cases} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{cases} \Rightarrow x = \frac{b_2c_1 - b_1c_2}{a_1b_2 - a_2b_1} \text{ and } y = \frac{a_1c_2 - a_2c_1}{a_1b_2 - a_2b_1}$$

Mathematicians defined the symbol

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$$

as a determinant of order 2 and the four numbers arranged in row and column were called its elements. If we write the coefficients of the equations in the following form

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$$

, then such an arrangement is called a

determinant. In a determinant, horizontal lines are known as rows and vertical lines are known as columns. The shape of every determinant is a square. If a determinant is of order n , then it contains n rows and n columns.

$$E. g. \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}, \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

are determinants of second and third order, respectively.

For every square matrix A of order $n \times n$, there exists a number associated with it called the determinant of a square matrix (<https://byjus.com/jee/matrices/>).

For a matrix of 1×1 , the determinant is $A = [a]$.

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$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \text{ the determinant is } ad - bc.$$

the value of the determinant is $= a(ei - fh) - b(di - fg) + c(dh - eg)$.

(i) The number of elements in a determinant of order n is n^2 .

Properties of Determinants

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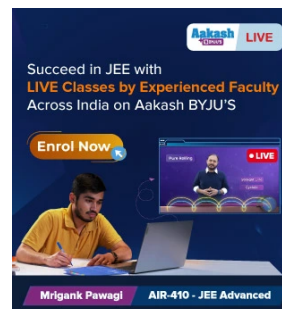
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$$\begin{vmatrix} 3 & 2 & 5 \\ 9 & -1 & 4 \\ 2 & 3 & -5 \end{vmatrix}$$

Solution:

$$\Delta = (a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32}) - (a_{13}a_{22}a_{31} + a_{11}a_{23}a_{32} + a_{12}a_{21}a_{33})$$

we can expand the given determinant.



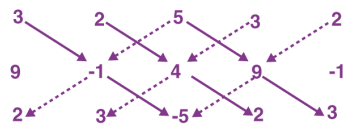
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Here,

$$\Delta = \begin{vmatrix} 3 & 2 & 5 \\ 9 & -1 & 4 \\ 2 & 3 & -5 \end{vmatrix} \Rightarrow \Delta = 15 - 36 + 90 + 16 + 1.$$

Illustration 2: Evaluate the determinant :

$$\begin{vmatrix} x^2 - x + 1 & x - 1 \\ x + 1 & x + 1 \end{vmatrix}$$

Solution:

By using the determinant expansion formula (<https://byjus.com/determinant-formula/>), we can get the result.

We have

$$\begin{vmatrix} x^2 - x + 1 & x - 1 \\ x + 1 & x + 1 \end{vmatrix} = (x^2 - x + 1)(x + 1) - (x - 1)(x + 1) \\ = x^3 + x^2 - x^2 - x + x + 1 - x^2 + 1 \\ = x^3 - x^2 + 2$$

Symmetric and Skew Symmetric Determinants

Symmetric Determinant

A determinant is called a **symmetric determinant** if $a_{ij} = a_{ji}, \forall i, j$

$$E. g. \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$$

Skew Symmetric Determinant

A determinant is called a **skew symmetric determinant** if $a_{ij} = -a_{ji}, \forall i, j$ for every element.

$$E. g. \begin{vmatrix} 0 & 3 & -1 \\ -3 & 0 & 5 \\ 1 & -5 & 0 \end{vmatrix}$$

Note: (i) $\det |A| = 0 \Rightarrow A$ is singular matrix (ii) $\det |A| \neq 0 \Rightarrow A$ is non-singular matrix

Multiplication of Two Determinants

(a) Multiplication of two second order determinants is as follows: (as R to C method)

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \times \begin{vmatrix} l_1 & m_1 \\ l_2 & m_2 \end{vmatrix} = \begin{vmatrix} a_1l_1 + b_1l_2 & a_1m_1 + b_1m_2 \\ a_2l_1 + b_2l_2 & a_2m_1 + b_2m_2 \end{vmatrix}$$

(b) Multiplication of two third order determinants is defined as follows

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \times \begin{vmatrix} l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \\ l_3 & m_3 & n_3 \end{vmatrix}$$

(as R to C method)

$$= \begin{vmatrix} a_1l_1 + b_1l_2 + c_1l_3 & a_1m_1 + b_1m_2 + c_1m_3 & a_1n_1 + b_1n_2 + c_1n_3 \\ a_2l_1 + b_2l_2 + c_2l_3 & a_2m_1 + b_2m_2 + c_2m_3 & a_2n_1 + b_2n_2 + c_2n_3 \\ a_3l_1 + b_3l_2 + c_3l_3 & a_3m_1 + b_3m_2 + c_3m_3 & a_3n_1 + b_3n_2 + c_3n_3 \end{vmatrix}$$

Note:

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(i) The two determinants to be multiplied must be of the same order.

(ii) To get the T_{mn} (term in the m^{th} row n^{th} column) in the product, Take the m^{th} row of the 1^{st} determinant and multiply it by the corresponding terms of the n^{th} column of the 2^{nd} determinant and add.

(iii) This method is the row-by-column multiplication rule for the product of 2 determinants of the n^{th} order determinant.

(iv) If Δ' is the determinant formed by replacing the elements of a Δ of order n with their corresponding co-factors, then

$\Delta' = \Delta^{n-1}$. Δ' is called the reciprocal determinant.

Illustration 3: Reduce the power of the determinant

$$\begin{vmatrix} 0 & c & b^2 \\ c & 0 & a \\ b & a & 0 \end{vmatrix}$$

to 1.

Solution:

By multiplying the given determinant two times, we get the determinant as required.

$$\begin{vmatrix} 0 & c & b^2 \\ c & 0 & a \\ b & a & 0 \end{vmatrix}^2 = \begin{vmatrix} 0 & c & b \\ c & 0 & a \\ b & a & 0 \end{vmatrix} \begin{vmatrix} 0 & c & b \\ c & 0 & a \\ b & a & 0 \end{vmatrix} \Rightarrow \begin{vmatrix} b^2 + c^2 & & \\ & ba & c^2 \\ & ca & \end{vmatrix}$$

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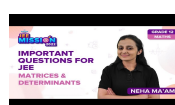
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Frequently Asked Questions

Q1 What is the determinant of an identity matrix?

The determinant of an identity matrix is 1.

Q2 Are determinants always positive?

No, determinants can be negative, positive or zero.

Q3 Give an application of determinants.

Determinants are used to solve linear equations in two or three variables.

Q4 What is the value of the determinant of a matrix, if any two rows or columns are identical?

If any two rows or columns of a matrix are identical, then the value of the determinant is equal to zero.

Q5 What will be the determinant of a matrix when two rows and columns are interchanged?

If two rows or columns of a determinant are interchanged, then the determinant changes its sign.

Q6 Will the determinant change if transposed?

No, the determinant of a matrix is equal to the determinant of the transpose of the matrix. $|A^T| = |A|$.

Q7 Does a matrix have more than one determinant?

No, a matrix cannot have more than one determinant.



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Q8 What happens to a determinant if all the elements of a row/column are multiplied by any scalar k ?

If all elements of a row/column of a determinant are multiplied by some scalar k , the value of the new determinant is k times the given determinant. If A is a square matrix having n rows, and k is any scalar, then $|kA| = k^n|A|$.

! Test your Knowledge on determinants



Put your understanding of this concept to test by answering a few MCQs. Click 'Start Quiz' to begin!

Select the correct answer and click on the "Finish" button
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